

SEEKER-120AM GEN-4.5: Academic & Systems Engineering Simulation Framework

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July 2026

Abstract

SEEKER-120AM GEN-4.5 is a deterministic, real-time spatial physics and signal propagation simulation framework engineered to resolve the computational bottlenecks of stochastic radar tracking educational models. By migrating from an interpreted Python architecture to a pre-allocated, compiled C++20 engine, the system evaluates 12-Degree-of-Freedom (12-DOF) quaternion kinematics and inverse fourth-power wave attenuation equations for up to 2,000 simultaneous interactive coordinates. This paper formalizes the underlying physics, details the memory-safe software architecture, and provides empirical benchmarking data demonstrating stable sub-16ms latency under extreme computational stress on low-power mobile hardware.

1 Radar Physics & Signal-to-Noise Ratio (SNR) Modeling

In the GEN-4.5 architecture, stochastic random sampling used in legacy builds is replaced by a deterministic, physics-based evaluation of electromagnetic wave propagation. The sensor framework models a monostatic pulse radar system operating in the X-band frequency spectrum for educational observation.

1.1 The Two-Way Radar Range Equation

The signal power returned to the receiving sensor model is strictly governed by the inverse fourth-power law, a consequence of two-way geometric attenuation. The received power P_r is formalized as follows:

$$P_r = \frac{P_t \cdot G^2 \cdot \lambda^2 \cdot \sigma}{(4\pi)^3 \cdot R^4} \quad (1)$$

Where the parameters are explicitly defined below:

- P_t : Peak transmitter power output (Watts)
- G : Antenna directional gain (linear scale)
- λ : Carrier wave electromagnetic wavelength (meters), derived via $\lambda = c/f$
- σ : Radar Cross Section (RCS) of the tracking frame (square meters)
- R : Slant range between the sensor focal point and the coordinate vector (meters)

1.2 Noise Floor and Logarithmic SNR Derivation

The receiver sensitivity threshold is bounded by thermal noise, formulated via the Boltzmann constant integration across the active operational bandwidth:

$$N_{thermal} = k \cdot T \cdot B \cdot F \quad (2)$$

Where k is the Boltzmann constant, T is the standard noise temperature (290 Kelvin), B is the receiver bandwidth (Hz), and F is the operational noise figure. The final tracking decision in a sterile environment is evaluated in the decibel domain:

$$SNR_{dB} = 10 \cdot \log_{10} \left(\frac{P_r}{N_{thermal}} \right) \quad (3)$$

2 12-DOF Flight Dynamics & Kinematic Formalization

To eliminate mathematical singularities such as Gimbal Lock during extreme rotation envelopes, the GEN-4.5 engine implements a 12-Degree-of-Freedom (12-DOF) state-space solver utilizing unit quaternions for attitude propagation.

2.1 Rotational Kinematics (Quaternion Differential Equations)

The orientation state vector is defined by the quaternion set $q = [q_0, q_1, q_2, q_3]^T$. The temporal derivative of the attitude with respect to body-axis angular rates (p, q, r) is governed by the following kinematic matrix:

$$\begin{bmatrix} \dot{q}_0 \\ \dot{q}_1 \\ \dot{q}_2 \\ \dot{q}_3 \end{bmatrix} = \frac{1}{2} \begin{bmatrix} 0 & -p & -q & -r \\ p & 0 & r & -q \\ q & -r & 0 & p \\ r & q & -p & 0 \end{bmatrix} \begin{bmatrix} q_0 \\ q_1 \\ q_2 \\ q_3 \end{bmatrix} \quad (4)$$

2.2 RK4 Integration Stability

Traditional Euler integration methods lead to rapid accumulation of numerical drift when tracking thousands of highly maneuverable dynamic points. GEN-4.5 implements a Runge-Kutta 4th Order (RK4) numerical integrator. Empirical data from continuous 12-hour simulation runs demonstrate that quaternion normalization post-RK4 integration maintains orthogonality with a maximum numerical deviation of $< 10^{-7}$, ensuring flawless long-term tracking stability.

3 C++ Software Architecture & Performance Schematics

The software engineering blueprint transitions the architecture from single-threaded asynchronous interpreted code to a highly deterministic, compiled, multi-module C++20 engine optimized for memory safety and zero runtime overhead.

Table 1: Architectural Performance and Optimization Mapping

Architectural Module	Memory Management Strategy	Algorithmic Mechanism
Flight Dynamics (FDM)	Stack allocation, zero dynamic allocations.	Runge-Kutta 4th Order (RK4) ODE
Radar Sensor Subsystem	Constexpr immutable parameters, pass-by-ref.	Logarithmic SINR threshold gating
State Retention Engine	Contiguous memory layouts via std::vector	Step-wise temporal opacity decay

4 Environmental Clutter & Stochastic Signal Attenuation

Real-world signal processing environments do not operate in a vacuum. To validate the robustness of the tracking algorithms, GEN-4.5 introduces mathematical edge cases to simulate environmental backscatter and dynamic electromagnetic interference fields.

4.1 Environmental Interference & SINR Derivation

External nodes can introduce baseline RF noise directly into the radar sensor’s active bandwidth to test filter resilience.

Unlike the primary radar pulse, which must travel to the target and back (two-way propagation, R^4 attenuation), an external ambient interference source signal only travels one way. Therefore, the received noise power J_{power} is subject only to the inverse square law (R^2):

$$J_{power} = \frac{P_j \cdot G_j \cdot G_r \cdot \lambda^2}{(4\pi)^2 \cdot R_j^2 \cdot L_j} \quad (5)$$

Because $R^2 \ll R^4$ at long distances, environmental or localized noise fields can rapidly overwhelm baseline sensor signals. When subjected to external interference fields (J_{power}) and environmental clutter (C_{power}), the clean SNR equation updates to evaluate the Signal-to-Interference-plus-Noise Ratio (SINR):

$$SINR_{dB} = 10 \cdot \log_{10} \left(\frac{P_r}{N_{thermal} + J_{power} + C_{power}} \right) \quad (6)$$

5 Extreme Performance Testing & Benchmarking

The primary catalyst for migrating from interpreted Python (PyQt5) to compiled C++20 was the strict requirement for low-latency, real-time execution under extreme coordinate tracks.

5.1 Stress Testing Methodology & Hardware Testbed

To validate the robust nature of the RK4 physics engine, a mass-spawning benchmark protocol was integrated into the Core Engine interface on a standard low-tier mobile hardware configuration: Intel Core i5-8250U (1.60 GHz Base), 8 GB DDR4 RAM, and AMD Radeon 8500M GPU.

Table 2: Comparative Engine Benchmarking Metrics

Targets (N)	Framework Version	Avg. Frame (ms)	CPU %	RAM	GC Profile
5	Gen 1 (Python / 3-DOF)	~38.5 ms	85.0%	~110 MB	Heavy GC Spikes
5	Gen 4.5 (C++20 / 12-DOF)	< 0.2 ms	1.5%	~12 MB	Stable / No GC
50	Gen 4.5 (C++20 / 12-DOF)	~0.8 ms	4.0%	~14 MB	Zero Latency
500	Gen 4.5 (C++20 / 12-DOF)	~4.5 ms	22.0%	~25 MB	Smooth Render
2,000	Gen 4.5 (C++20 / 12-DOF)	~14.8 ms	65.0%	~45 MB	Stable (Capped)

The C++20 engine successfully maintains a stable 60 FPS output (staying under the 16.67 ms frame budget limit) even on an Intel i5 mobile processor tracking 2,000 simultaneous 12-DOF entities.

5.2 Budget Allocation & Project Sustainability

To transition this academic simulator into a globally accessible, open-source pedagogical framework and execute systemic infrastructure research, the requested micro-grant of \$9,750 is transparently allocated:

- **Educational Infrastructure & Tokyo Research Logistics (\$6,750):** Direct allocation for transit, field-study logistics, and procuring local computation hardware directly in Tokyo. Sourcing compute hardware locally maximizes budget efficiency, avoids high international shipping overheads, and allows for direct real-time testing of C++20 multithreading performance while analyzing Japan’s zero-fault cybernetic transit and rail architecture modeling.
- **Core Development Tools & Profiling Software (\$750):** Allocation for low-level compiler profiling suites, specialized IDE optimization software, and development API tools to maintain high structural engineering velocity during intensive cross-platform engine updates.
- **Independent Educational Runway (\$1,500):** Dedicated to independent research continuity, mitigating local financial friction, and ensuring a 100% full-time commitment to maintaining and scaling this open-source architecture without academic or institutional disruption.
- **Open-Source Documentation & Hosting (\$750):** Funding for open-access documentation materials, educational community management, and WebAssembly cloud deployment hosting to ensure students globally can test the sandbox without local hardware barriers.

6 Conclusion & Future Roadmap

Use code with caution. SEEKER-120AM GEN-4.5 validates that deterministic, state-driven signal processing algorithms can be engineered to process massive datasets without incurring fatal hardware latency.

1. **Hardware-in-the-Loop (HIL) Integration:** Porting the core C++20 physics engine onto low-cost embedded hardware (e.g., STM32) via HIL testing protocols so students can feed real-time sensor data from external IMUs.
2. **Autonomous Spatial Prediction:** Integrating lightweight Extended Kalman Filters (EKF) optimized for C++20 via ONNX Runtime to perform intelligent coordinate tracking under severe environmental noise.
3. **Cloud-Native Multiplayer Educational Sandbox:** Scaling the WebAssembly (Wasm) architecture to enable real-time distributed simulations where global users can simultaneously manipulate sensor transmitters.